

COURSE: CHE 312: Transport Phenomena II

Spring Semester 2020

Department of Chemical Engineering

University of Illinois at Chicago

Midsemester Exam #3

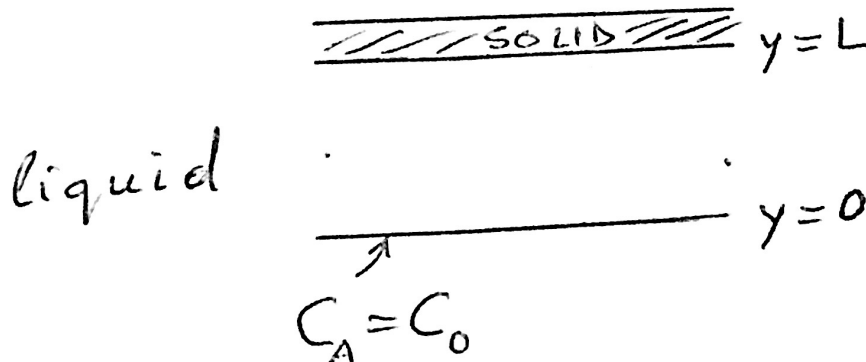
Online

Duration: 50 minutes --- April 13, 2020

Problem 1: (50 points)

Consider the irreversible conversion of species A into B, C and F through the reactions $A \rightarrow B$, $A \rightarrow C$, and $A \rightarrow F$ in a stationary liquid film of thickness L . Determine C_A throughout the film, if the surface at $y = L$ is assumed to be impermeable and unreactive, and the concentration of A at $y = 0$ is specified as C_0 . Two homogeneous reactions follow zeroth order kinetics with rates of reaction $R_{AB} = -k_B$, $R_{AC} = -k_C$; the third homogeneous reaction follows first order kinetics with rate of reaction $R_{AF} = -k_F \cdot C_A$.

(Bonus 10 points) Find the overall rate of reactant consumption.



Problem 2: (50 points)

Consider steady state diffusion of the species A through two media arranged as shown in the attached diagram. Assume that diffusion is one-dimensional. At $x = -L$, $C_{A1} = C_L$, and at $x = L$, $C_{A2} = C_R$. In medium 2, there is a zeroth order chemical reaction through which the species A is consumed. Develop an expression for the concentration profiles across the two media.

(Bonus 10 points) Develop an expression for the steady state flux across the two media

